

E1

Consider the circuit in Fig. 1. The circuit parameters at $T = 25^\circ\text{C}$ are the following:

$V_{CC} = 12\text{ V}$; $R_{G1} = 56\text{ k}\Omega$; $R_{G2} = 27\text{ k}\Omega$; $R_{B1} = 68\text{ k}\Omega$; $R_{B2} = 56\text{ k}\Omega$; $R_{C1} = 8.2\text{ k}\Omega$; $R_{S2} = 200\text{ }\Omega$;
 $R_F = 60\text{ k}\Omega$; $R_S = 10\text{ k}\Omega$; $C_S = 100\text{ nF}$; $C_B = 1\text{ }\mu\text{F}$; $C_F = 1\text{ }\mu\text{F}$; $C_{CB} = 22\text{ pF}$.

Q_1 (NPN BJT): $V_{BE} = 0.7\text{ V}$; $\beta_F = \beta_0 = 200$; $V_{CEsat} = 0.2\text{ V}$; $r_0 = +\infty$;

Q_2 (E-NMOS): $V_{th} = 2\text{ V}$; $I_{DSS} = k_n'(W/2L)V_{th}^2 = 1\text{ mA}$; $r_0 = +\infty$;

$V_T = 26\text{ mV}$.

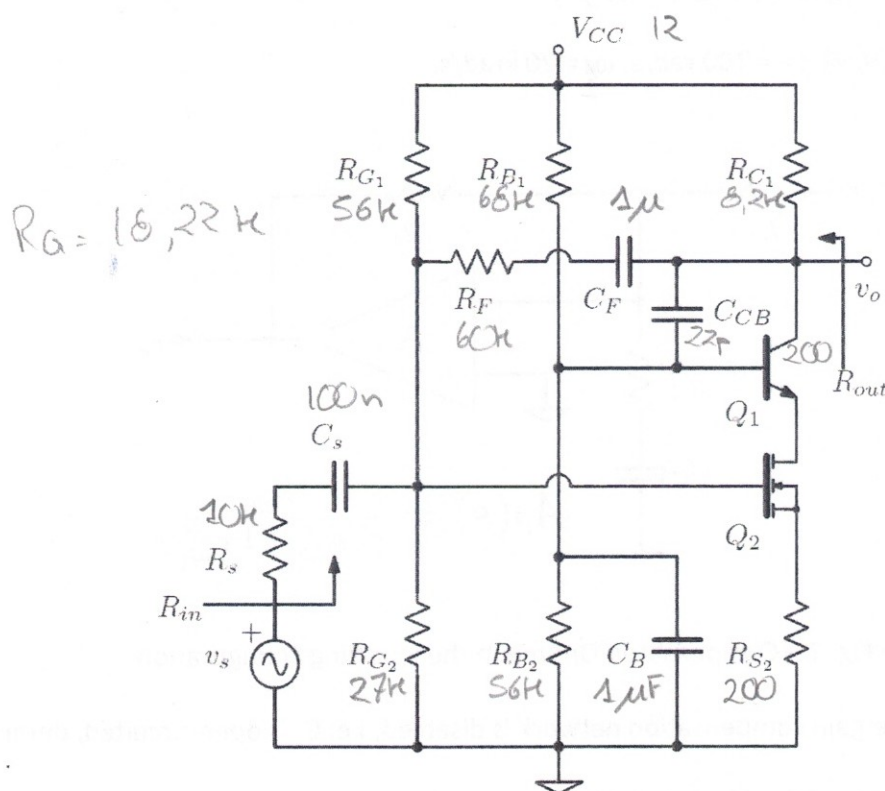


Fig. 1 – A case of two-stage feedback amplifier.

Considering *all capacitors* to be equivalent to *open circuits*, determine:

1. the operating points (V_{CE} , I_C) and (V_{DS} , I_D) of Q_1 and Q_2 ;
2. the quiescent power dissipation of Q_1 and Q_2 ;

Assuming that, at the frequencies of interest (i.e. at mid-band), capacitors C_S , C_B and C_F are equivalent to *short circuits*, while C_{CB} is equivalent to an *open circuit*, determine also:

3. the voltage gain $A_v = v_o/v_s$ of the amplifier;
4. the input resistance R_{in} indicated in the figure;
5. the output resistance R_{out} indicated in the figure.

Finally, considering the effect of capacitor C_{CB} only, determine

6. the *high frequency* bandwidth limit f_H of the amplifier.

E2

Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R_1 = 33 \text{ k}\Omega$; $R_2 = 82 \text{ k}\Omega$; $R_C = 33 \text{ }\Omega$; $C_C = 22 \text{ }\mu\text{F}$.

OpAmp: $A_{d0} = 10^5 \text{ [V/V]}$, $\omega_0 = 100 \text{ rad/s}$, $\omega_1 = 20 \text{ krad/s}$.

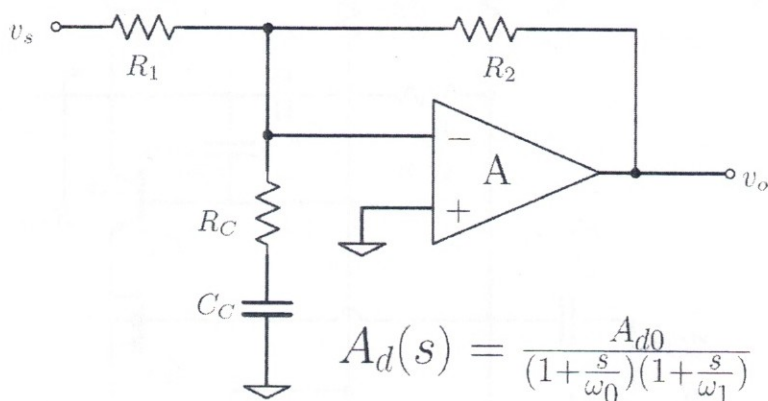


Fig. 2 – Compensated OpAmp in the inverting configuration.

Assuming the noise-gain compensation network is disabled, i.e. C_C is open circuited, determine:

1. the expression of the *ideal* circuit voltage gain $A_v(s) = v_o/v_s$;
2. the amplifier bandwidth and *phase margin*.

Then, after activating the compensation network, calculate also:

3. the new amplifier bandwidth and *phase margin*;
4. the expression of the *ideal* circuit voltage gain $A_v(s) = v_o/v_s$.